Sheet 1

Q2. a) $g : \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right] \to \left[\frac{3\pi}{4} + \frac{1}{\sqrt{2}}, \frac{5\pi}{4} - \frac{\sqrt{2}}{2}\right]$

The range is a subset of the domain ✓

$g'(x) = 1 + \cos x$

in $\left[\frac{3\pi}{4}, \frac{5\pi}{4}\right]$, $|g'(x)| < 1$ ✓ g is a contraction

So Banach's CMT applies ✓

b) $g : [0, 2] \to [-3, 17]$

$g'(x) = 3x^2 + 6x$

on $[0, 2]$ $g'(x) \in [0, 24]$ so $|g'(x)| > 1$

not a contraction X

c) $X = [1, \infty)$

$g(x) = x - x^{-1}$

$g'(x) = 1 + \frac{1}{x^2}$

on $[1, \infty)$, $g'(x) \in [1, 2]$

so $|g'(x)| > 1$ at $x = 1$

so not a contraction X

Q3. a) $e_{n+1} \approx M e_n^{\,q}$

Given e_n, e_{n+1}, e_{n+2}

$$\overset{\shortparallel}{M e_n^{\,q}} \qquad \overset{\shortparallel}{M (e_{n+1})^q} = M^{q+1} e_n^{\,2q}$$

$$\left(\frac{e_{n+1}}{e_n}\right)^q = \left(\frac{e_{n+2}}{e_{n+1}}\right)$$

$$q \log\left(\frac{e_{n+1}}{e_n}\right) = \log\left(\frac{e_{n+2}}{e_{n+1}}\right)$$

$$q = \frac{\log\left(\frac{e_{n+2}}{e_{n+1}}\right)}{\log\left(\frac{e_{n+1}}{e_n}\right)}$$

Then $M = \dfrac{e_{n+1}}{e_n^{\,q}}$